THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 11: **Electricity** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Two resistors of 4 ohm and 12 ohm in series are connected to a 16 V battery. The current is 1

(a) 4 A **(b)** 16 A **(c)** 1 A **(d)** 0.25 A

2. Heat produced by 3 A passing through 4 ohm for 10 s is 1

(a) 12 J **(b)** 90 J **(c)** 360 J **(d)** 120 J

3. If the potential difference across a fixed resistor is doubled, the current becomes 1

(a) twice **(b)** half **(c)** four times **(d)** unchanged

4. A circuit has 5 ohm in series with the parallel combination of 6 ohm and 3 ohm. The total resistance is 1

(a) 14 ohm **(b)** 9 ohm **(c)** 7 ohm **(d)** 3.5 ohm

5. If the current through a resistor is doubled, the heat produced in a given time becomes 1

(a) half **(b)** twice **(c)** four times **(d)** unchanged

6. For a graph between V (y-axis) and I (x-axis), the quantity represented by the slope is 1

(a) resistance **(b)** power **(c)** charge **(d)** conductance

7. A 6 V battery is connected to two resistors of 3 ohm and 6 ohm in series. The potential difference across the 3 ohm resistor is 1

(a) 2 V **(b)** 3 V **(c)** 6 V **(d)** 4 V

8. A fuse rated 5 A can safely run appliances up to a total power at 220 V of 1

(a) 5500 W **(b)** 1100 W **(c)** 220 W **(d)** 44 W

9. The resistance of a wire is 10 ohm. If its length is tripled by stretching with the same volume, the resistance is nearly 1

(a) 3.3 ohm **(b)** 30 ohm **(c)** 10 ohm **(d)** 90 ohm

10. Two identical bulbs are connected in series to a battery. Compared to a single bulb, each bulb is 1

(a) off **(b)** dimmer **(c)** equally bright **(d)** brighter

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the power of a heater that draws 4 A from a 220 V supply. 2

12. Find the cost of using a 60 W bulb for 5 hours daily for 30 days at Rs 6 per unit. 2

13. Calculate the resistance of 100 m of copper wire of area 1 mm^2 if rho = 1.7×10^{-8} ohm m. 2

14. The heat produced in a resistor is 400 J in 10 s when the current is 2 A. Find the resistance. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A wire has a resistance of 16 ohm. It is cut into four equal parts and the parts are joined in parallel. Find the equivalent resistance. 3

16. Compare the resistivity of copper (1.6×10^{-8}) and nichrome (1.1×10^{-6}) ohm m. For wires of the same size, find the ratio of resistances. 3

17. Why is a fuse wire of a lower melting point and a smaller rating than the wire of the appliance? 3

OR

In a series circuit of 5 ohm and 10 ohm with a 6 V source, find the potential difference across each and the power dissipated in the circuit. 3

18. A 2 ohm resistor is connected in series with a parallel combination of 3 ohm and 6 ohm and a 6 V battery. Find the current from the battery. 3

OR

How much charge passes through a bulb of 100 W at 220 V in 10 minutes? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A wire of length 2 m, area 0.4 mm^2 and resistivity 1.6×10^{-8} ohm m is connected to a 4 V battery. Find its resistance, the current and the power dissipated. 5

OR

A battery of 12 V and a bulb of 6 V, 3 W is to be used with a resistor so that the bulb glows normally. Find the series resistor. 5

20. Derive the formula for the equivalent resistance of two resistors in parallel and find the equivalent of 8 ohm and 24 ohm and the current from 12 V. 5

OR

Explain why the resistance of a wire changes when it is stretched to double its length. Find the new resistance if the original is 5 ohm. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND